4-yearly maintenance

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Master data

Serial number	822883	**Plant**	S-03855-004
Type	E-82 E2	**Location**	Ger
Control type	CS82a		

Order

Order type	4-yearly maintenance
Order number	95748289
Confirmation date	Apr 23, 2019
Report number	940000834965
Document number	TD-esc-08-xxx-xxx-15-055 Rev007

Inspection report

Summary

Without findings	267	**With findings**	6	**Total**	273

Details

No.	Work step	Result
	Electrics - Tower base	
1	Check tower base warning systems	1 Done - No defects
2	Updating software	1 Done - No defects
3	Checking the WEC parameters and operating parameters	1 Done - No defects
4	Checking the rechargeable battery of the ENERCON SCADA Server UPS	3 Not done - Not necessary
5	Check access control	3 Not done - Not necessary
6	Checking the shadow shutdown light sensors	3 Not done - Not necessary
7	Measuring earth resistance	1 Done - No defects
8	Measuring rotor blade de-icing currents	3 Not done - Not necessary
	Electrics - External transformer station	
9	Checking sockets in external transformer station	3 Not done - Not necessary
10	Measuring insulation resistance on sockets in external transformer station	3 Not done - Not necessary
11	Measuring the earth resistance - external transformer station	3 Not done - Not necessary
12	Checking capacitive voltage taps in external transformer station	3 Not done - Not necessary
13	Checking transformer room in external transformer station	3 Not done - Not necessary
14	Visually checking transformer in external transformer station	3 Not done - Not necessary
15	Taking an oil sample in external transformer station	3 Not done - Not necessary

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No.	Work step	Result
16	Checking medium-voltage switchgear in external transformer station	3 Not done - Not necessary
17	Checking medium-voltage switchgear (every four years) in external transformer station	3 Not done - Not necessary
18	Checking low-voltage distribution system in external transformer sta- tion	3 Not done - Not necessary
19	Checking terminal lugs of low-voltage distribution system - external transformer station	3 Not done - Not necessary
20	Checking station basement in external transformer station	3 Not done - Not necessary
	Electrics - WEC medium voltage area	
21	Checking the capacitive voltage taps	1 Done - No defects
22	Checking the transformer room	1 Done - No defects
23	Carrying out a visual inspection of the transformer	1 Done - No defects
24	Taking and oil sample	3 Not done - Not necessary
25	Checking medium-voltage switchgear	3 Not done - Not necessary
26	Checking the medium-voltage switchgear every 4 years	1 Done - No defects
27	Checking the low-voltage distribution system	3 Not done - Not necessary
28	Checking LV distribution system fans	3 Not done - Not necessary
29	Checking low-voltage distribution system terminal lugs	3 Not done - Not necessary
30	Checking the terminal lugs for the secondary systems cabinet	3 Not done - Not necessary
31	Checking control cabinet terminal lugs	1 Done - No defects
32	Checking UPS terminal lugs	1 Done - No defects
33	Checking NH1 fuse switch disconnector terminal lugs	3 Not done - Not necessary
34	Checking the foundation basement	3 Not done - Not necessary
35	Checking AC busbar system V3	3 Not done - Not necessary
	Electrics - Tower base	
36	Measuring continuity of protective conductor system	1 Done - No defects
37	Measuring insulation resistance	1 Done - No defects
38	Checking electrical cabinets	1 Done - No defects
39	Checking the electrical cabinets overvoltage protection	1 Done - No defects
40	Checking the SCADA interface backup battery	1 Done - No defects
	Electrics - External transformer station	
41	Measuring continuity on protective conductor system in external trans- former station	3 Not done - Not necessary
42	Performing electrical check on RCD in external transformer station	3 Not done - Not necessary
	Electrics - WEC medium voltage area	
43	Checking the ENERCON transformer protection system	1 Done - No defects
	Electrics - Tower base	
44	Checking the tower cooling system	1 Done - No defects

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45	Checking the UPS	1 Done - No defects
46	Checking DC charger unit	3 Not done - Not necessary
47	Checking UPS heating	1 Done - No defects
48	Checking the heating system for the power cabinets and control cab- inet	1 Done - No defects
49	Checking the power cabinet fans	1 Done - No defects
50	Testing emergency lights	1 Done - No defects
51	Performing an electrical inspection on the RCD	1 Done - No defects
52	Performing an electrical inspection on an upstream RCD	3 Not done - Not necessary
53	Measuring short circuit current	1 Done - No defects
54	Checking the auxiliary contacts	1 Done - No defects
55	Checking the LV earth fault detection system current sensor	3 Not done - Not necessary
56	Checking the service hoist emergency lighting system	2 Done - Defects detected
57	Performing an electrical inspection on the service hoist RCD	1 Done - No defects
58	Checking sockets	1 Done - No defects
59	Measuring insulation resistance of sockets	1 Done - No defects
	Electrics - Tower	
60	Checking sockets	1 Done - No defects
61	Measuring insulation resistance of sockets	1 Done - No defects
62	Performing an electrical inspection on the RCD	1 Done - No defects
	Electrics - Machine house	
63	Nacelle warning systems	1 Done - No defects
64	Checking the horns	1 Done - No defects
65	Checking the telephone connection	1 Done - No defects
66	Nacelle orientation to north and cable twisting	1 Done - No defects
67	Checking the rotor lock	1 Done - No defects
68	Checking rotor blade de-icing system	3 Not done - Not necessary
69	Performing an electrical functional check on the winch	1 Done - No defects
70	Checking sockets	1 Done - No defects
71	Measuring insulation resistance of sockets	1 Done - No defects
72	Testing emergency lights	1 Done - No defects
73	Performing an electrical inspection on the RCD	1 Done - No defects
74	Checking the yaw drives	1 Done - No defects
75	Checking the smoke detectors	1 Done - No defects
76	Checking the nacelle noise sensor	3 Not done - Not necessary
77	Checking the nacelle control cabinet fan	3 Not done - Not necessary

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78	Checking the nacelle control cabinet heating	1 Done - No defects
79	Checking the excitation controller fan	1 Done - No defects
80	Checking heater in excitation controller box	3 Not done - Not necessary
81	Checking the yaw inverter cabinet heating	1 Done - No defects
82	Checking the generator filter cabinet fan	1 Done - No defects
83	Checking the generator filter cabinet terminal lugs	1 Done - No defects
84	Checking the rectifier fan	1 Done - No defects
85	Checking the rectifier heating	1 Done - No defects
86	Checking the auxiliary contacts	1 Done - No defects
87	Checking the cable twist limit switch	3 Not done - Not necessary
88	Checking cam limit switch	1 Done - No defects
89	Checking electrical cabinets	1 Done - No defects
90	Checking the electrical cabinets overvoltage protection	1 Done - No defects
91	Measuring short circuit current	1 Done - No defects
92	Measuring continuity of protective conductor system	1 Done - No defects
93	Testing torque monitoring system	1 Done - No defects
94	Polarisation index test	1 Done - No defects
	Electrics - Rotor head	
95	Testing lights	1 Done - No defects
96	Checking electrical cabinets	1 Done - No defects
97	Checking the electrical cabinets overvoltage protection	1 Done - No defects
98	Measuring short circuit current	1 Done - No defects
99	Measuring continuity of protective conductor system	1 Done - No defects
100	Checking sockets	1 Done - No defects
101	Measuring insulation resistance of sockets	1 Done - No defects
102	Performing an electrical inspection on the RCD	1 Done - No defects
103	Checking the Cobham slip ring unit and Label slip ring unit	1 Done - No defects
104	Checking the ENERCON slip ring unit	3 Not done - Not necessary
105	Checking the torque support and drive mechanism	1 Done - No defects
106	Checking the generator	1 Done - No defects
107	Checking rotor monitoring system	1 Done - No defects
108	Checking rotor head noise sensor	1 Done - No defects
109	Checking the air gap switch	1 Done - No defects
110	Checking rotor blade A’s limit switch	1 Done - No defects
111	Checking rotor blade B’s limit switch	1 Done - No defects

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112	Checking rotor blade C's limit switch	1 Done - No defects
113	Checking pitch motor fans of rotor blade A	1 Done - No defects
114	Checking pitch motor fans of rotor blade B	1 Done - No defects
115	Checking pitch motor fans of rotor blade C	1 Done - No defects
116	Checking the rotor head discharge box	1 Done - No defects
117	Checking discharge box of rotor blade A	1 Done - No defects
118	Checking discharge box of rotor blade B	1 Done - No defects
119	Checking discharge box of rotor blade C	1 Done - No defects
	Electrics - Attachment parts outside the	
120	Checking the ultrasonic anemometer alignment	1 Done - No defects
121	Check precipitation sensor	3 Not done - Not necessary
122	Checking the beacon system	1 Done - No defects
	Electrics - Machine house	
123	Checking air cooling system	1 Done - No defects
124	Checking the rectifier earth fault monitoring system	1 Done - No defects
125	Checking generator residual current monitoring	1 Done - No defects
	Electrics - Emergency shutdowns	
126	Checking the nacelle control cabinet EMERGENCY STOP buttons	1 Done - No defects
127	Testing the control cabinet EMERGENCY STOP button	1 Done - No defects
128	Testing the tower entrance door EMERGENCY STOP button	1 Done - No defects
129	Testing the E-module EMERGENCY STOP button	3 Not done - Not necessary
130	Checking the main safety circuit relay K4 emergency shutdown	1 Done - No defects
131	Checking the main safety circuit relay K3 emergency shutdown	1 Done - No defects
132	Checking the rotor lock emergency shutdown	1 Done - No defects
133	Checking the vibration sensor emergency shutdown	1 Done - No defects
	Electrics - Machine house	
134	Measuring rotor blade de-icing currents	3 Not done - Not necessary
	Mechanical - Tower base	
135	Visual inspection of tower base	1 Done - No defects
136	Checking foundation	1 Done - No defects
137	Checking bolt connections on bolt cage with load ring	3 Not done - Not necessary
138	Checking fire extinguisher in tower base	1 Done - No defects
	Mechanical - Tower	
139	Checking the tower anchorage points	1 Done - No defects
140	Checking the tower ventilation system	1 Done - No defects

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141	Checking the tower bolt connections	1 Done - No defects
142	Checking the concrete tower	1 Done - No defects
143	Checking the prestressing system	3 Not done - Not necessary
144	Checking steel tower	3 Not done - Not necessary
145	Checking steel section	1 Done - No defects
146	Visual inspection of tower	1 Done - No defects
	Mechanical - Safety ladder	
147	Checking the safety ladder stickers	1 Done - No defects
148	Checking start and end of climbing route	1 Done - No defects
149	Check C-profile and fall arrest noses	1 Done - No defects
150	Checking fastening brackets and clamps	1 Done - No defects
151	Checking rungs	1 Done - No defects
152	Checking gap widths on ladder sections	1 Done - No defects
153	Visually inspecting safety ladder	1 Done - No defects
154	Checking safety catches	1 Done - No defects
155	Checking the lateral rail exits	1 Done - No defects
156	Checking the small rest platforms	1 Done - No defects
157	Checking the safety ladder fasteners	1 Done - No defects
158	Performing a test climb	1 Done - No defects
159	Applying “Next Check” inspection label	1 Done - No defects
	Mechanical - Service hoist	
160	Checking that the document pouch is complete	1 Done - No defects
161	Checking the safety signs	1 Done - No defects
162	Checking the lift cage	1 Done - No defects
163	Checking the anchorage points	1 Done - No defects
164	Checking the cable guides	1 Done - No defects
165	Checking the key-operated switch in the lift cage	1 Done - No defects
166	Checking the safety guard and cable pulley in the lift cage	1 Done - No defects
167	Checking the lift cage crossbeam	1 Done - No defects
168	Checking the door safety switch	1 Done - No defects
169	Checking the top limit switch	1 Done - No defects
170	Checking the bottom limit switches	1 Done - No defects
171	Checking the bottom outside sensor	1 Done - No defects
172	Checking the emergency stop button	1 Done - No defects
173	Checking the emergency control unit	1 Done - No defects

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174	Checking the hoist cable and safety cable at the hoist touchdown platform	1 Done - No defects
175	Checking the cable fasteners with landing box	3 Not done - Not necessary
176	Checking the cable fasteners without landing box	1 Done - No defects
177	Checking the tension of the guide cables	1 Done - No defects
178	Checking the lighting in the lift cage	1 Done - No defects
179	Checking the indicator lights	1 Done - No defects
180	Checking service hoist winch	1 Done - No defects
181	Checking the safety catch	1 Done - No defects
182	Check automatic operation	4 Not done - Assembly suspended
183	Checking the steel cables	1 Done - No defects
184	Oiling the wire ropes	1 Done - No defects
185	Checking the cable guides at the landing platforms	1 Done - No defects
186	Checking the trailing cable and the restraint	1 Done - No defects
187	Checking the landing platform switches on the platforms	3 Not done - Not necessary
188	Checking the guardrail locking device	1 Done - No defects
189	Checking the top limit switches and the cut-off device	1 Done - No defects
190	Checking the upper tower crossbeam	1 Done - No defects
191	Performing a test run	1 Done - No defects
192	Performing the finishing steps on the service lift	1 Done - No defects
	Mechanical - Machine house	
193	Checking the machine house anchorage points	1 Done - No defects
194	Checking chain winch	1 Done - No defects
195	Checking the machine house first aid kit	1 Done - No defects
196	Checking the machine house fire extinguisher	1 Done - No defects
197	Checking the Firespy fire-extinguishing system in the machine house	3 Not done - Not necessary
198	Checking the machine house fitted parts	1 Done - No defects
199	Checking the machine house casing	1 Done - No defects
200	Checking the machine house hatches	1 Done - No defects
201	Checking the main carrier and the axle pin	1 Done - No defects
202	Checking yaw gears	1 Done - No defects
203	Checking the yaw bearing	1 Done - No defects
204	Checking the stator shield	1 Done - No defects
205	Testing rotor lock	1 Done - No defects
206	Checking the electromechanical brake	1 Done - No defects
207	Checking the shaft sleeve	1 Done - No defects

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208	Checking the machine house bolt connections	1 Done - No defects
209	Visual inspection of machine house	1 Done - No defects
	Mechanical - Rotor head	
210	Checking the rotor head anchorage points	1 Done - No defects
211	Taking a front and rear main bearing grease sample	1 Done - No defects
212	Greasing the rear and front rotor bearing	1 Done - No defects
213	Checking the rotor	1 Done - No defects
214	Checking the rotor head casing	1 Done - No defects
215	Checking the rotor hub and the blade adapters	1 Done - No defects
216	Checking rotor head seals	1 Done - No defects
217	Checking central lubrication system on rotor head	2 Done - Defects detected
218	Checking pitch gear	1 Done - No defects
219	Checking blade flange bearing	1 Done - No defects
220	Check lightning protection system of rotor blade A	1 Done - No defects
221	Check lightning protection system of rotor blade B	1 Done - No defects
222	Check lightning protection system of rotor blade C	1 Done - No defects
223	Checking the rotor head bolt connections	1 Done - No defects
224	Visual inspection of rotor head	1 Done - No defects
	Mechanical - Rotor blade	
225	Checking the rotor blade A bolt connections	1 Done - No defects
226	Checking the rotor blade B bolt connections	1 Done - No defects
227	Checking the rotor blade C bolt connections	1 Done - No defects
228	Checking the TES spring bolts on rotor blade A	1 Done - No defects
229	Checking the TES spring bolts on rotor blade B	1 Done - No defects
230	Checking the TES spring bolts on rotor blade C	1 Done - No defects
231	Check rotor blade A base plate	2 Done - Defects detected
232	Check rotor blade B base plate	2 Done - Defects detected
233	Check rotor blade C base plate	2 Done - Defects detected
234	Check laminate on rotor blade A	1 Done - No defects
235	Check laminate on rotor blade B	1 Done - No defects
236	Check laminate on rotor blade C	1 Done - No defects
237	Check web bonding on rotor blade A	1 Done - No defects
238	Check web bonding on rotor blade B	1 Done - No defects
239	Check web bonding on rotor blade C	1 Done - No defects
240	Check crescent on rotor blade A	1 Done - No defects

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241	Check crescent on rotor blade B	1 Done - No defects
242	Check crescent on rotor blade C	1 Done - No defects
243	Check the connection between the lightning protection cable and the discharge ring on rotor blade A	1 Done - No defects
244	Check the connection between the lightning protection cable and the discharge ring on rotor blade B	1 Done - No defects
245	Check the connection between the lightning protection cable and the discharge ring on rotor blade C	1 Done - No defects
246	Check lightning protection cable on rotor blade A	1 Done - No defects
247	Check lightning protection cable on rotor blade B	1 Done - No defects
248	Check lightning protection cable on rotor blade C	1 Done - No defects
249	Check the connection between the lightning protection cable and the flat profile on rotor blade A	1 Done - No defects
250	Check the connection between the lightning protection cable and the flat profile on rotor blade B	1 Done - No defects
251	Check the connection between the lightning protection cable and the flat profile on rotor blade C	1 Done - No defects
252	Check flat profile of web on rotor blade A	1 Done - No defects
253	Check flat profile of web on rotor blade B	1 Done - No defects
254	Check flat profile of web on rotor blade C	1 Done - No defects
255	Checking the exterior of rotor blade A	1 Done - No defects
256	Checking rotor blade B exterior	1 Done - No defects
257	Checking rotor blade C exterior	1 Done - No defects
258	Check flange spoiler longitudinal bolts for rotor blade A	1 Done - No defects
259	Check flange spoiler longitudinal bolts for rotor blade B	1 Done - No defects
260	Check flange spoiler longitudinal bolts for rotor blade C	1 Done - No defects
	Mechanical - Nacelle exterior	
261	Checking anchorage points nacelle exterior	1 Done - No defects
262	Checking fitted parts nacelle exterior	1 Done - No defects
	country-specific maintenance items	
263	Checking rescue barrel	2 Done - Defects detected
264	Check the safety ladder safety cable	3 Not done - Not necessary
	Finishing steps	
265	Performing paintwork repairs	3 Not done - Not necessary
266	Acoustic check of WEC	1 Done - No defects
	Measuring device	
1000	Type of measuring device	Fluke 1654b
2000	ID number	1147048,0
3000	Calibrated	on

4-yearly maintenance

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4000	Date - month	10,0
5000	Date - year	2018,0
	Measured values for item 7	
10	Air humidity	normal
20	Soil condition	humid
30	Outside temperature	9,0 °C
40	Measurement 1	9,70 OHM
50	Measurement 2	9,30 OHM
60	Measurement 3	8,60 OHM
70	Average value	9,20 OHM
80	Orientation of measuring probes	G
90	Test set-up with decoiler, 50m (SAP 146596)	No
	Measured values for item 8	
	Measured values for item 10	
	Measured values for item 11	
	Measured values for item 36	
10	Bonding bar - foundation earth electrode	0,070 OHM
20	Bonding bar - tower entrance stairs	0,050 OHM
30	Bonding bar - crescent housing lighting for external area of tower door	0,250 OHM
40	Bonding bar - tower door frame (concrete towers only)	0,120 OHM
50	Bonding bar - Tower door lighting housing, indoor	0,090 OHM
60	Bonding bar - tower door socket	0,130 OHM
70	Bonding bar - tower door industrial power socket	0,150 OHM
80	Bonding bar - safety ladder	0,050 OHM
90	Bonding bar - service hoist socket	0,270 OHM
110	Bonding bar - service hoist housing	0,250 OHM
190	Bonding bar - 2nd bonding bar	0,040 OHM
200	Bonding bar - transformer	0,030 OHM
210	Bonding bar - grease pan of transformer	0,090 OHM
220	Bonding bar - switchgear of transformer	0,050 OHM
230	Bonding bar - remote control unit of MV switchgear =164	0,110 OHM
240	Bonding bar - Motor housing for tower fan 1	0,020 OHM
250	Bonding bar - Motor housing for tower fan 2	0,060 OHM
360	Bonding bar - control cabinet =005	0,060 OHM
370	Bonding bar - cabinet lighting socket in control cabinet =005	0,090 OHM

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390	Bonding bar - UPS cabinet =068	0,010 OHM
400	Bonding bar - socket in UPS cabinet =069	0,090 OHM
420	Bonding bar - power cabinet =006+1	0,020 OHM
430	Bonding bar - power cabinet =006+2	0,020 OHM
440	Bonding bar - power cabinet =006+3	0,010 OHM
450	Bonding bar - power cabinet =006+4	0,010 OHM
460	Bonding bar - power cabinet =006+5	0,020 OHM
470	Bonding bar - power cabinet =006+6	0,020 OHM
480	Bonding bar - power cabinet =006+7	0,010 OHM
	Measured values for item 37	
10	Phase L1 - against PE of power cabinet 1	500,00 MOH
20	Phase L2 - against PE of power cabinet 1	500,00 MOH
30	Phase L3 - against PE of power cabinet 1	500,00 MOH
40	Phase L1 - against PE of power cabinet 2	500,00 MOH
50	Phase L2 - against PE of power cabinet 2	500,00 MOH
60	Phase L3 - against PE of power cabinet 2	500,00 MOH
70	Phase L1 - against PE of power cabinet 3	500,00 MOH
80	Phase L2 - against PE of power cabinet 3	500,00 MOH
90	Phase L3 - against PE of power cabinet 3	500,00 MOH
100	Phase L1 - against PE of power cabinet 4	500,00 MOH
110	Phase L2 - against PE of power cabinet 4	500,00 MOH
120	Phase L3 - against PE of power cabinet 4	500,00 MOH
130	Phase L1 - against PE of power cabinet 5	500,00 MOH
140	Phase L2 - against PE of power cabinet 5	500,00 MOH
150	Phase L3 - against PE of power cabinet 5	500,00 MOH
160	Phase L1 - against PE of power cabinet 6	500,00 MOH
170	Phase L2 - against PE of power cabinet 6	500,00 MOH
180	Phase L3 - against PE of power cabinet 6	500,00 MOH
190	Phase L1 - against PE of power cabinet 7	500,00 MOH
200	Phase L2 - against PE of power cabinet 7	500,00 MOH
210	Phase L3 - against PE of power cabinet 7	500,00 MOH
	Measured values for item 41	
	Measured values for item 42	
	Measured values for item 51	
10	Socket next to entrance door, phase angle 0°	36,10 MSC

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20	Socket next to entrance door, phase angle 0°	0,0 V
30	Socket next to entrance door, phase angle 180°	27,10 MSC
40	Socket next to entrance door, phase angle 180°	0,0 V
50	Socket next to entrance door, phase angle 0°	27,00 MA
60	Socket next to entrance door, phase angle 0°	0,0 V
70	Socket next to entrance door, phase angle 180°	27,00 MA
80	Socket next to entrance door, phase angle 180°	0,0 V
	Measured values for item 52	
	Measured values for item 53	
10	Power cabinet 1 output L1	1700,0 A
20	Power cabinet 1 output L2	2500,0 A
30	Power cabinet 1 output L3	2300,0 A
40	Control cabinet =005 output line contactor L1	2100,0 A
50	Control cabinet =005 output line contactor L2	2500,0 A
60	Control cabinet =005 output line contactor L3	2100,0 A
	Measured values for item 57	
10	Service hoist socket, phase angle 0°	35,70 MSC
20	Service hoist socket, phase angle 0°	0,0 V
30	Service hoist socket, phase angle 180°	27,50 MSC
40	Service hoist socket, phase angle 180°	0,0 V
50	Service hoist socket, phase angle 0°	24,00 MA
60	Service hoist socket, phase angle 0°	0,0 V
70	Service hoist socket, phase angle 180°	24,00 MA
80	Service hoist socket, phase angle 180°	0,0 V
	Measured values for item 59	
10	Phase L1 industrial power socket, tower door - PE	200,00 MOH
20	Phase L2 industrial power socket, tower door - PE	200,00 MOH
30	Phase L3 industrial power socket, tower door - PE	200,00 MOH
40	Neutral conductor industrial power socket, tower door - PE	200,00 MOH
50	Phase L1 against phase L2 industrial power socket, tower door - PE	200,00 MOH
60	Phase L1 against phase L3 industrial power socket, tower door - PE	200,00 MOH
70	Phase L2 against phase L3 industrial power socket, tower door - PE	200,00 MOH
80	Phase of socket, tower door - PE	200,00 MOH
90	Neutral conductor of socket, tower door - PE	200,00 MOH
120	Phase of socket, in service hoist - PE	200,00 MOH

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130	Neutral conductor of socket in service hoist - PE	200,00 MOH
200	Phase of cabinet lighting socket in control cabinet =005 - PE	200,00 MOH
210	Neutral conductor of cabinet lighting socket in control cabinet - PE	200,00 MOH
220	Socket phase in UPS cabinet =068 - PE	200,00 MOH
230	Socket, neutral wire in UPS cabinet =068 - PE	0,00 MOH
	Measured values for item 61	
10	Phase of upper socket in tower - PE	200,00 MOH
20	Neutral conductor of upper socket in tower - PE	115,00 MOH
30	Phase L1 of upper industrial power socket in tower - PE	200,00 MOH
40	Phase L2 of upper industrial power socket in tower - PE	200,00 MOH
50	Phase L3 of upper industrial power socket in tower - PE	200,00 MOH
60	Neutral conductor of upper industrial power socket in tower - PE	200,00 MOH
70	Phase L1 against phase L2 of upper industrial power socket in tower	200,00 MOH
80	Phase L1 against phase L3 of upper industrial power socket in tower	200,00 MOH
90	Phase L2 against phase L3 of upper industrial power socket in tower	200,00 MOH
	Measured values for item 62	
10	Topmost socket in tower, 0°	36,50 MSC
20	Topmost socket in tower, 0°	0,0 V
30	Topmost socket in tower, 180°	28,50 MSC
40	Topmost socket in tower, 180°	0,0 V
50	Top socket in tower, 0°	27,00 MA
60	Top socket in tower, 0°	0,0 V
70	Top socket in tower, 180°	24,00 MA
80	Top socket in tower, 180°	0,0 V
	Measured values for item 71	
10	Phase L1 of industrial power socket 1 on nacelle control cabinet - PE	200,00 MOH
20	Phase L2 of industrial power socket 1 on nacelle control cabinet - PE	200,00 MOH
30	Phase L3 of industrial power socket 1 on nacelle control cabinet - PE	200,00 MOH
40	Neutral conductor of industrial power socket 1 nacelle control cabinet - PE	200,00 MOH
50	Phase L1 against L2 of industrial power socket 1 on nacelle control cabinet	200,00 MOH
60	Phase L1 against L3 of industrial power socket 1 on nacelle control cabinet	200,00 MOH
70	Phase L2 against L3 of industrial power socket 1 on nacelle control	200,00 MOH
80	Phase of exterior tower lighting switch socket - PE	200,00 MOH
90	Neutral conductor of exterior tower lighting switch socket - PE	200,00 MOH
100	Phase of socket on nacelle control cabinet - PE	200,00 MOH

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110	Neutral conductor of socket on nacelle control cabinet - PE	200,00 MOH
120	Phase of socket in main carrier - PE	200,00 MOH
130	Neutral conductor of socket in main carrier - PE	50,00 MOH
	Measured values for item 73	
10	Socket on nacelle control cabinet, phase angle 0°	36,70 MSC
20	Socket on nacelle control cabinet, phase angle 0°	0,0 V
30	Socket on nacelle control cabinet, phase angle 180°	26,90 MSC
40	Socket on nacelle control cabinet, phase angle 180°	1,0 V
50	Socket on nacelle control cabinet, phase angle 0°	24,00 MA
60	Socket on nacelle control cabinet, phase angle 0°	1,0 V
70	Socket on nacelle control cabinet, phase angle 180°	24,00 MA
80	Socket on nacelle control cabinet, phase angle 180°	1,0 V
	Measured values for item 91	
10	Nacelle control cabinet, input nacelle power supply L1	430,0 A
20	Nacelle control cabinet, input nacelle power supply L2	445,0 A
30	Nacelle control cabinet, input nacelle power supply L3	1200,0 A
	Measured values for item 92	
20	Bonding bar - industrial power socket on main carrier	0,930 OHM
30	Bonding bar - socket in main carrier	0,950 OHM
40	Bonding bar - exterior tower lighting switch socket	0,080 OHM
50	Bonding bar - excitation controller box =018	0,000 OHM
60	Bonding bar - fan, frequency converter cabinet =088+1	0,000 OHM
70	Bonding bar - Yaw inverter cabinet = 102	0,010 OHM
80	Bonding bar - nacelle control cabinet =015	0,000 OHM
90	Bonding bar - industrial power socket 1 on nacelle control cabinet	0,030 OHM
100	Bonding bar - socket on nacelle control cabinet	0,020 OHM
130	Bonding bar - winch motor housing	0,190 OHM
140	Bonding bar - beacon system control cabinet=115	0,030 OHM
150	Bonding bar - yaw motor 1	0,010 OHM
160	Bonding bar - yaw motor 2	0,000 OHM
170	Bonding bar - yaw motor 3	0,030 OHM
180	Bonding bar - yaw motor 4	0,020 OHM
190	Bonding bar - yaw motor 5	0,050 OHM
200	Bonding bar - yaw motor 6	0,020 OHM
210	Bonding bar - rectifier cabinet =017+1	0,000 OHM

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220	Bonding bar - rectifier cabinet =017+2	0,020 OHM
230	Bonding bar - filter cabinet =066 part 1.1	0,000 OHM
240	Bonding bar - filter cabinet =066 part 1.2	0,010 OHM
250	Bonding bar - filter cabinet =066 part 2.1	0,040 OHM
260	Bonding bar - filter cabinet =066 part 2.2	0,030 OHM
270	Bonding bar - Machine house casing (only for aluminium nacelle casing)	0,010 OHM
280	Bonding bar - stator sub distribution system =014	0,010 OHM
290	Bonding bar - stator	0,080 OHM
300	Bonding bar - rotor (brake disk)	0,040 OHM
310	Bonding bar - rotor brake =147+A	0,030 OHM
320	Bonding bar - rotor brake =147+B	0,020 OHM
330	Bonding bar - motor housing for generator air 1	0,030 OHM
340	Bonding bar - motor housing for generator air 2	0,110 OHM
350	Bonding bar - motor housing for generator air 3	0,120 OHM
360	Bonding bar - motor housing for generator air 4	0,020 OHM
370	Bonding bar - motor housing for generator air 5	0,090 OHM
380	Bonding bar - motor housing for generator air 6	0,050 OHM
390	Bonding bar - surge protection device, generator =019	0,040 OHM
400	Bonding bar - residual current monitoring, generator =148	0,030 OHM
410	Bonding bar - anemometer suppor frame 1	0,030 OHM
	Measured values for item 94	
10	Air humidity	73,00 %
20	Stator temperature	40,00 °C
30	Rotor temperature	42,00 °C
40	Unit =017 single phase from System 1 to PE (1min)	390,000 MOH
50	Unit =017 single phase from System 1 to PE (10min)	440,000 MOH
60	Unit =017 single phase from System 1 to PE [PI]	1,130
70	Unit =017 single phase from System 2 to PE (1min)	440,000 MOH
80	Unit =017 single phase from System 2 to PE (10min)	530,000 MOH
90	Unit =017 single phase from System 2 to PE [PI]	1,200
100	Unit =017 single phase from System 1 to System 2 (1min)	327,000 MOH
110	Unit =017 single phase from System 1 to System 2 (10min)	352,000 MOH
120	Unit =017 single phase from System 1 to System 2 [PI]	1,080
	Measured values for item 98	
10	Rotor sub-distribution system =013 L1 -FC04:1	240,0 A

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	Measured values for item 99	
10	Bonding bar - spinner lighting	0,020 OHM
20	Bonding bar - blade relay box A =011	0,000 OHM
30	bonding bar - pitch control box A =012	0,090 OHM
40	Bonding bar - compact limit switch A =028	0,010 OHM
60	Bonding bar - pitch motor 1 A =053	0,000 OHM
70	Bonding bar - load control box =075+A (LoadControl)	0,010 OHM
80	Bonding bar - rotor blade A capacitor box =076	0,050 OHM
90	Bonding bar - blade relay box B =011	0,030 OHM
100	Bonding bar - pitch control box B =012	0,030 OHM
110	Bonding bar - compact limit switch B =028	0,000 OHM
120	Bonding bar - pitch motor 1 B =053	0,010 OHM
130	Bonding bar - load control box =075+B (LoadControl)	0,010 OHM
140	Bonding bar - rotor blade B capacitor box =076	0,020 OHM
150	Bonding bar - blade relay box C =011	0,020 OHM
160	Bonding bar - pitch control box C =012	0,020 OHM
170	Bonding bar - compact limit switch C =028	0,010 OHM
180	Bonding bar - pitch motor 1 C =053	0,010 OHM
190	Bonding bar - load control box =075+C (LoadControl)	0,020 OHM
200	Bonding bar - rotor blade C capacitor box =076	0,010 OHM
210	Bonding bar - rotor sub-distribution system =013	0,040 OHM
220	Bonding bar - socket on the rotor	0,040 OHM
230	Bonding bar - hub sub-distribution system =035	0,010 OHM
240	Bonding bar - slip ring unit =051	0,000 OHM
250	Bonding bar - transformer cabinet =133	0,000 OHM
260	Bonding bar - hub	0,050 OHM
270	Bonding bar - rotor head casing (with PE cable connected)	0,040 OHM
280	Bonding bar - earthing cable - rotor head	0,030 OHM
	Measured values for item 101	
10	Phase of socket, on the rotor - PE	200,00 MOH
20	Neutral conductor of socket on the rotor - PE	200,00 MOH
	Measured values for item 102	
10	Socket on the rotor, phase angle 0°	35,70 MSC
20	Socket on the rotor, phase angle 0°	0,0 V
30	Socket on the rotor, phase angle 180°	26,70 MSC

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40	Socket on the rotor, phase angle 180°	1,0 V
50	Socket on the rotor, phase angle 0°	24,00 MA
60	Socket on the rotor, phase angle 0°	1,0 V
70	Socket on the rotor, phase angle 180°	24,00 MA
80	Socket on the rotor, phase angle 180°	1,0 V
	Measured values for item 134	
	Measured values for item 160	
10	next (third-party) inspection - month	1,0
20	next (third-party) inspection - year	2020,0
	Measured values for item 180	
10	Winch	Other type
20	Operating hours	4,0 H
30	YOM of winch	2017,0
	Measured values for item 194	
10	YOM of winch	2008,0

Findings / Notification overview

No.	Defect location	Defect type	Notification
56	tower, service hoist - cage, emergency lights	special entries - defect type not found	920035526858
95	rotor system - lighting, spinner	special, lighting - illuminant defect	920028890712
216	rotor system - central lubrication, hoses	special, sealing - sealing leakage	920035624115
217	rotor system - central lubrication, pump	special, grease system - piston distributor damage	920035363852
231	blade system A - bearing, inspection opening	material, metal - damage	920035975962
232	blade system B - bearing, inspection opening	material, metal - damage	920035976963
233	blade system C - bearing, inspection opening	material, metal - damage	920035976964
263	notice for next maintenance - expiration date fire ext./first-aid kit	material, metal - damage	920035976965

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